

Visual Evidence Degradation

Target-Object Masking

Ratios 25% · 50% · 75% · 100%

Temporal schedules

Occluded → Visible	■	■	■	■
Visible → Occluded	■	■	■	■
Before Grasp	■	■	■	■
After Grasp	■	■	■	■

Target-Receptacle Masking

Ratios 25% · 50% · 75% · 100%

Temporal schedules

Occluded → Visible	■	■	■	■
Visible → Occluded	■	■	■	■
Before Placement	■	■	■	■
After Placement	■	■	■	■

Interaction-Cue Masking

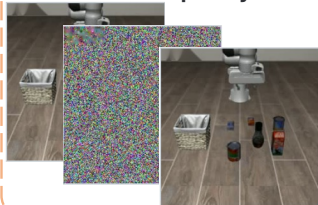


Gripper

object contact

Placement contact

Temporary Observation Corruption



Corruption: Noise / Blackout

Events: 1× / 2× / 3×

Phase: from approach to pre-placement(five)

View Unavailability

Third-person or Wrist

single-view

copy view

black fill

white fill



12 challenge categories

96 experimental settings

3,296 task-condition cases

Camera Staleness

Mechanism

delay

Freeze

Replay

Delay(replans)

$\Delta 1$

$\Delta 3$

$\Delta 5$

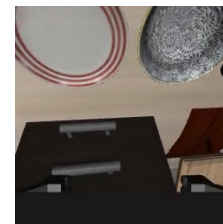
Affected Camera

Third-person

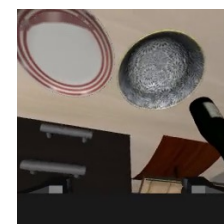
Wrist

Fixed Delay · Different View ($\Delta = 1$)

Current Wrist(t)

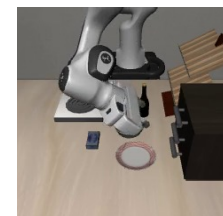


Input Wrist (from t-1)



≠

Current Third-person (t)



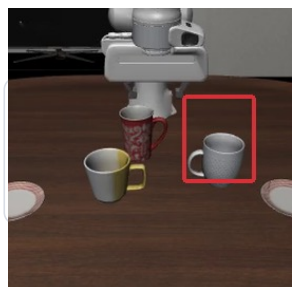
Input Third-person (from t-1)



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Visual Source Consistency

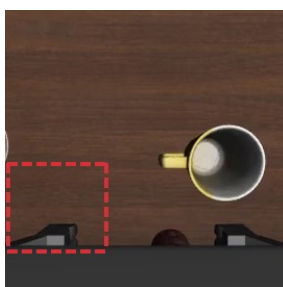
Cross-View Contradiction



Third-person truthful

Quantity

Position



Wrist Contradictory

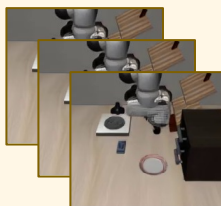
Existence

State

Prediction-Reality Divergence



t0: Input real observation



t1~tn: Input predicted Frames



Model

Task-Relevant Scene Variation

Task-Relevant Distractors



Shape

Category

Affordance

Color

Position

Trajectory

Target Geometry Variation

Deformation

Taper

Multipeak

Twist

Asymmetric

Severity

Low

Medium

High

Task-Precondition Variation



Receptacle Openness

Object pose

Visual State Ambiguity



Receptacle Openness

Object pose